

ARTIFICIAL MILLIMETER-WAVE GYROTROPIC STRUCTURE

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ABSTRACT. - A gyrotropic structure consisting of two gratings of wire helices embedded into a dielectric plate is proposed and investigated in the millimeter-wave band. This structure possesses the following characteristic property. At the operating frequency, the electric vector of the transmitted wave is rotated to an angle of 90° with respect to the electric vector of the incident wave, and this rotation angle is independent of the orientation of the latter vector.

INTRODUCTION. The gratings composed of wire helices represent examples of chiral electrodynamics structures [1]. In [2], a method for solving the diffraction problem for a plane wave normally incident on a single grating of wire helices was developed.

This method is based on the substitution of a cylinder with anisotropic surface conductivity along helical lines for a real multifilar helix. The authors of [3] theoretically demonstrated the realization of the gyrotropic effect for two gratings situated in a homogeneous space. Some experimental results for gratings composed of wire helices were presented in [4, 5]. In this paper, we analyze the gyrotropic properties of two cascaded gratings embedded into a dielectric plate.

DESIGN. The structure to be analyzed is shown in Fig. 1. It consists of two identical gratings of wire helices. The gratings are rotated with respect to each other to an angle of 45° about the axis and are symmetrically arranged in the dielectric plate of thickness $2D = 8.4$ mm. For reasons of simple technology, each of the gratings is embedded into a dielectric plate of thickness $D = 4.2$ mm. The grating period is $p = 2.2$ mm. The elements of the gratings are quadrifilar helices fabricated by winding a copper wire of diameter 0.1 mm around a nylon thread of diameter $2a = 1.2$ mm. The pitch of each conductor is 2.2 mm, the twirl angle between the wire axis and the nylon-thread axis being $\psi \approx 62^\circ$. The role of the dielectric is to reinforce the construction. As the dielectric, we chose an epoxy material whose dielectric permittivity is $\epsilon = 3 - j0.1$ in the frequency band considered. A detailed description of the fabrication technology of such gratings was presented in [5].

THEORY. Suppose that two gratings are situated in a homogeneous medium with the permittivity ϵ and the following conditions are fulfilled [3]: - the gratings are rotated with respect to each other to an angle of 45° about the axis, - the twirl angle of the helical lines is $\psi \approx 62^\circ$, - the distance between the gratings d and their period p are related by $d \approx 1.9p$, - the filling factor of the gratings is $2a/p \approx 0.59$. Then, at the frequency satisfying the condition $p/\lambda \approx 0.64$, where λ is the wavelength in the homogeneous medium with the dielectric permittivity ϵ , the cascade of two gratings transmits without reflection the waves both with the left-hand and right-hand polarizations and provides a phase shift of 180° between these waves. This means that the electric vector of a linearly polarized wave transmitted through the structure rotates to an angle of 90° irrespective of the orientation of the electric vector of the incident wave. In contrast to [3], there exist two dielectric-air interfaces in the structure under consideration. In order to avoid the reflection loss due to these interfaces, the thickness of the dielectric layer is chosen so that the transmission resonance occurs at the operating frequency.

EXPERIMENT. A sample of dimension 110 x 130 mm was situated between two horns. The transmitting horn radiates a linearly polarized field. The receiving horn and the sample can rotate. Figures 2-3 represent the measured characteristics of the transmitted wave at an operating frequency of 47.5 GHz. These figures demonstrate the transmission coefficient (Fig. 2), the axial ratio (Fig. 2), and the angle of polarization-plane rotation (Fig. 3) versus the rotation angle of the sample. It can be seen that the axial ratio of the transmitted field is no worse than 0.26 in the whole range of angles. The transmitted power ripple does not exceed 1.5 dB. The rotation angle of the polarization plane somewhat differs from the calculated value equal to 90° , the rotation angle ripple being no greater than 10° . These insignificant deviations from the isotropic behavior could be due to technological errors, thermal loss in the dielectric, as well as the inadequateness of the analyzed model. To a certain extent, the range properties of the structures under consideration are illustrated by the curves in Fig. 4. This figure represents the transmitted power as a function of frequency for four different orientations of the sample chosen with a step of 45° .

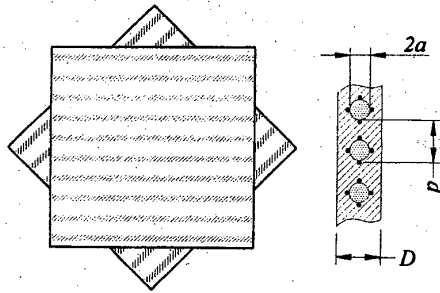


Fig. 1

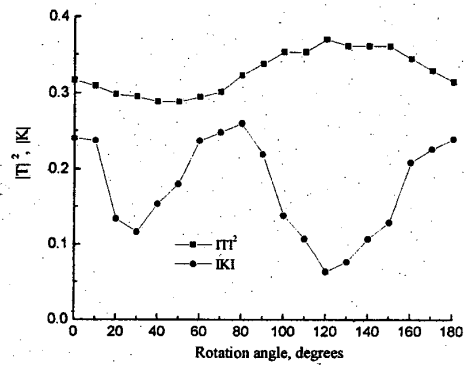


Fig. 2

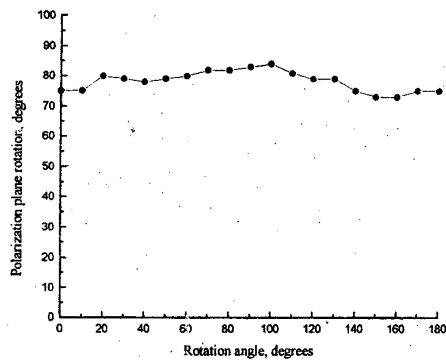


Fig. 3

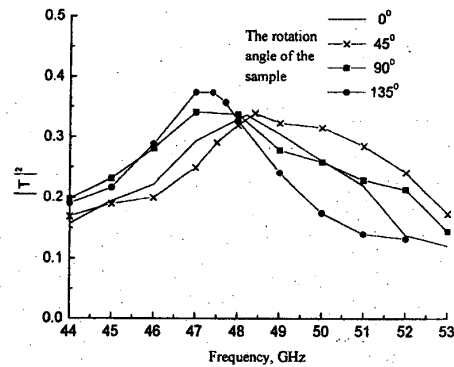


Fig. 4

CONCLUSIONS. An isotropic millimeter-wave transmission polarization rotator is proposed and designed. This device consists of two gratings made of multifilar wire helices embedded into a dielectric plate.

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